MANCHESTER 1824
The University of Manchester

Understanding the Earth Workshop

Week 5 Earthquakes

Learning objectives

In this workshop we will:

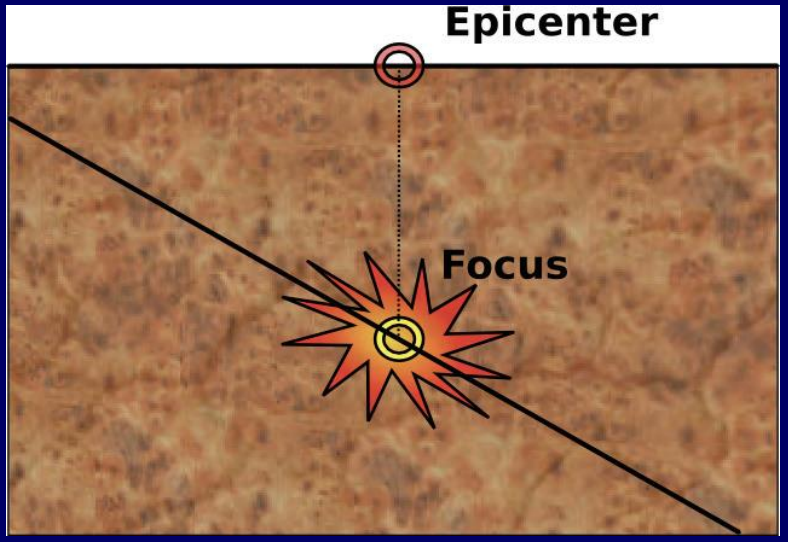
- compare and contrast the different types of seismic waves generated by earthquakes
- identify P and S waves generated by an earthquake on a seismograph
- locate an earthquake epicenter on a map using calculations made of the arrival times of P and S waves
- identify some natural hazards associated with earthquakes and assess the risk for different locations
- elucidate the relationship between the oil and gas exploration and seismicity

16:09:37 on 28 March 2005 N. Sumatra 8.7 M_w earthquake



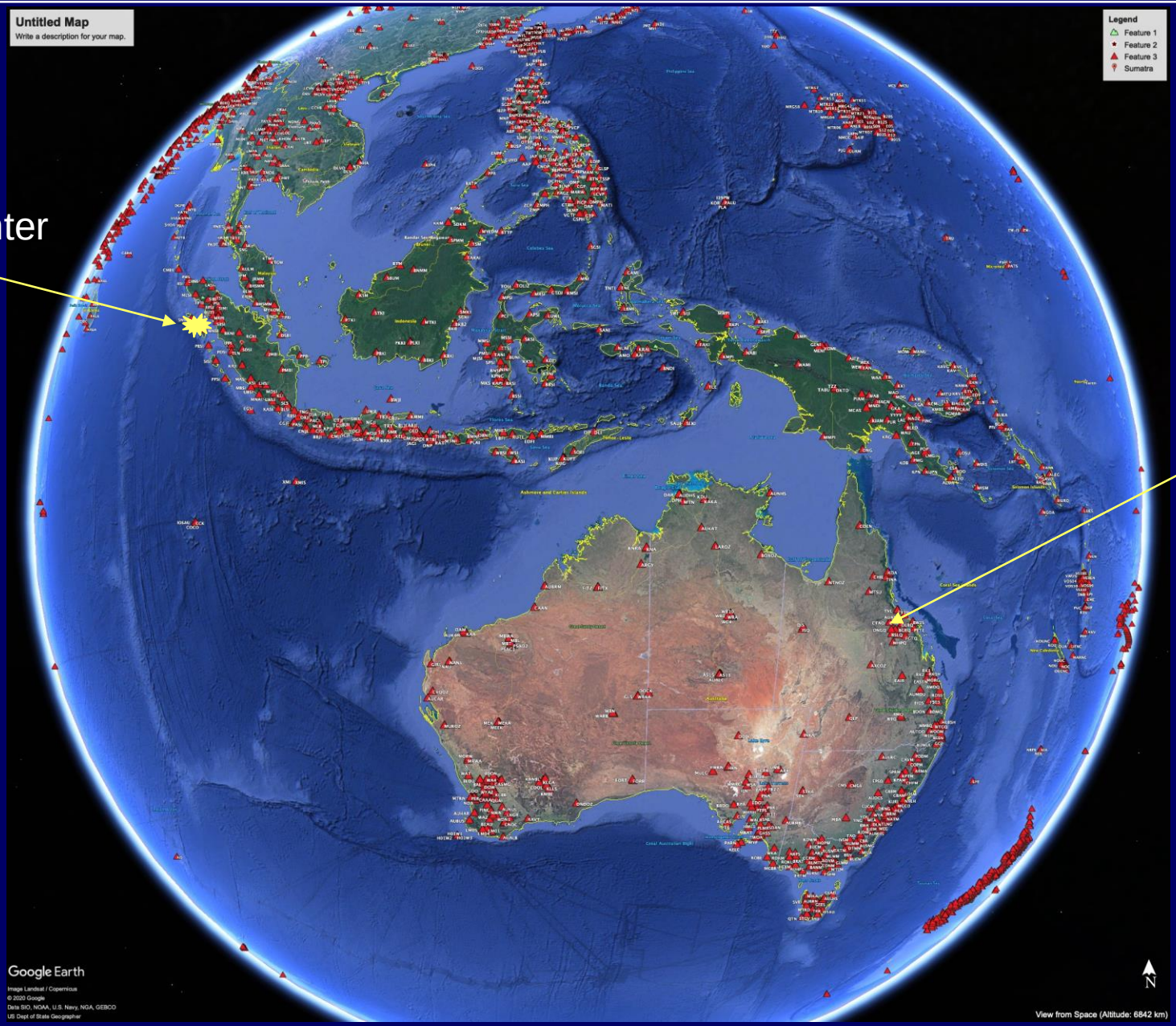
Main shock M8.7

- 135 km from Sibolga
- 490 km from Kuala Lumpur
- 300 km from Medan
- 1260 km from Jakarta

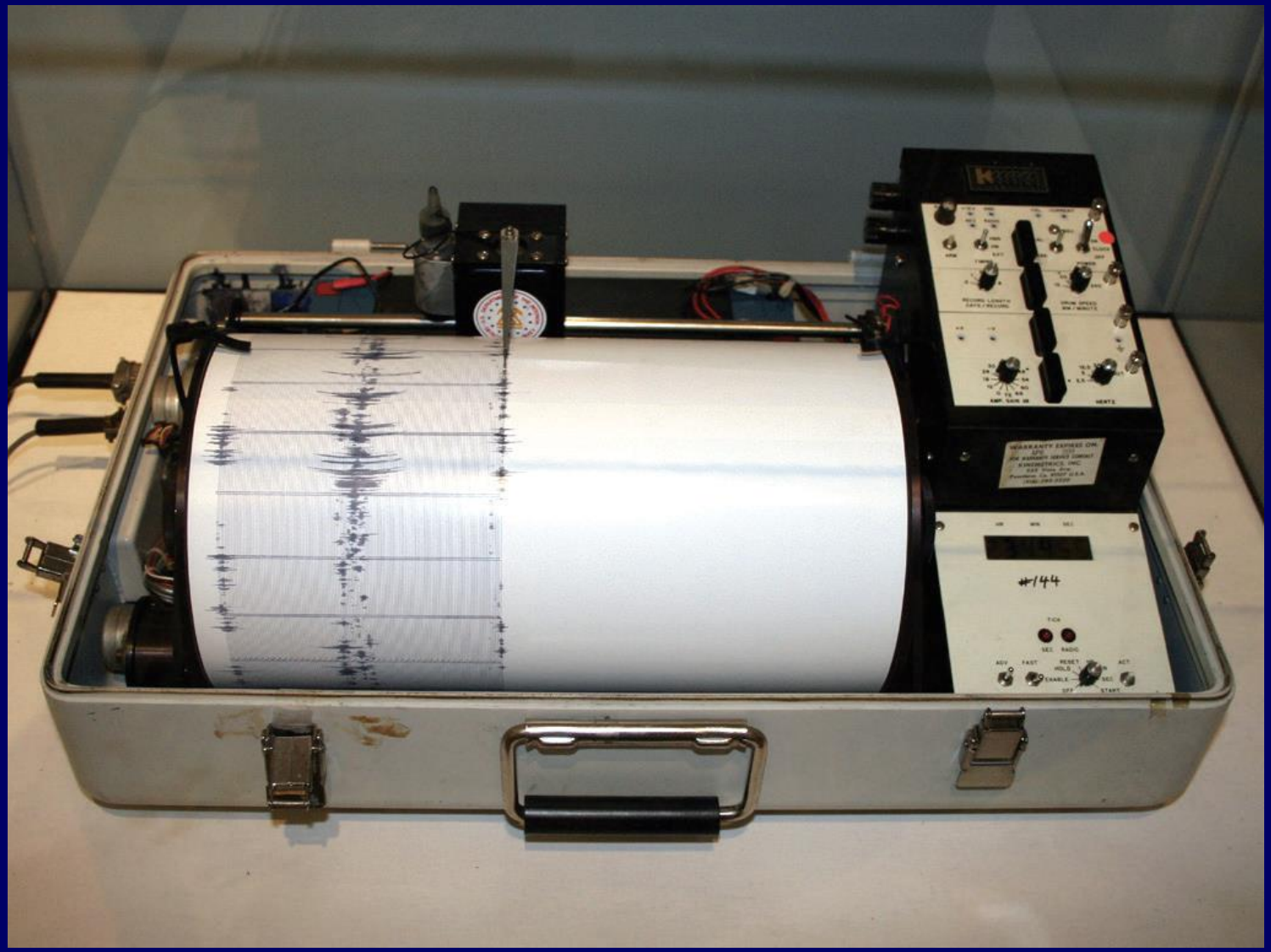




Global seismograph stations

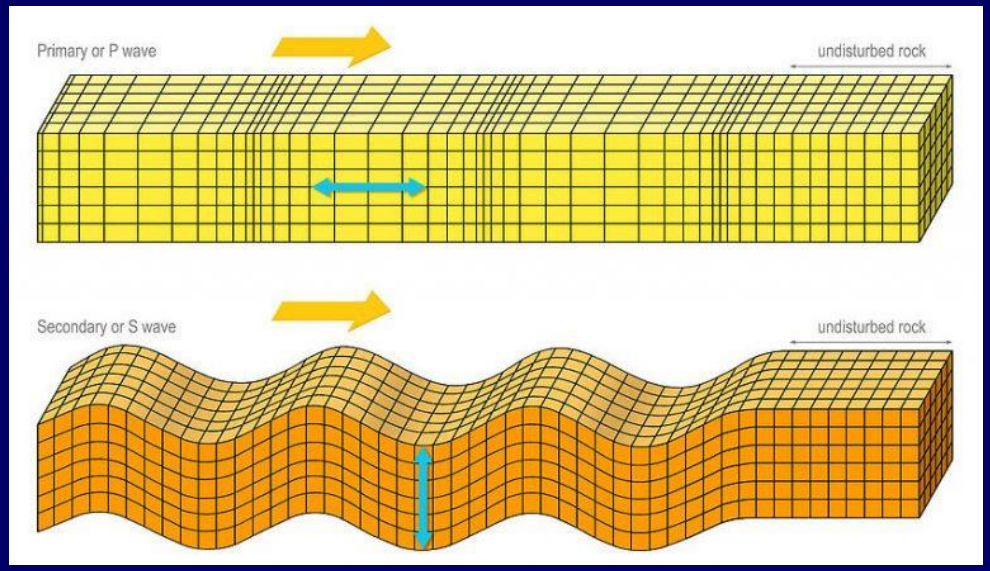


Seismology



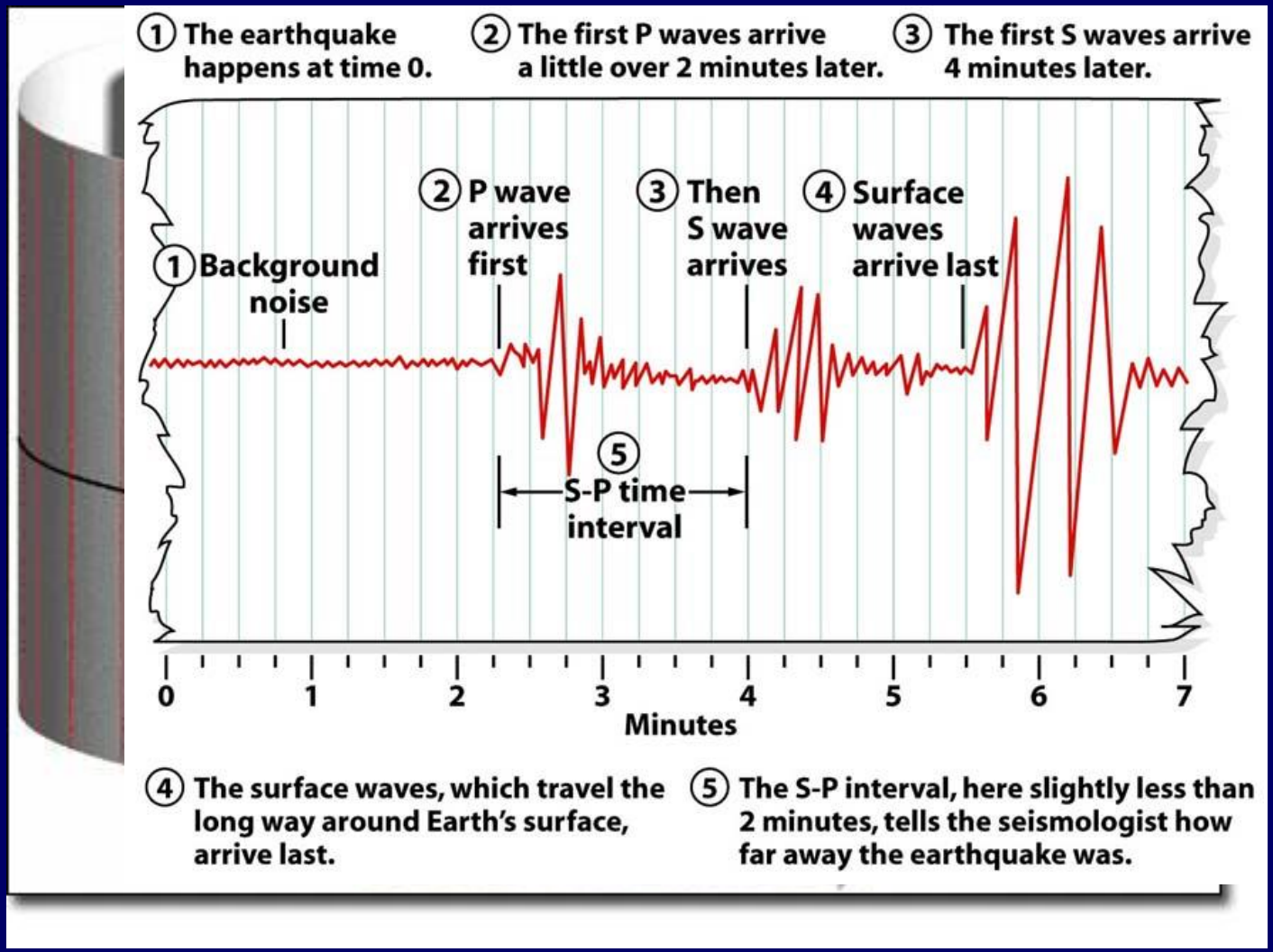
P and S waves

- Seismic waves are elastic waves generated by earthquakes
- We are interested in body waves i.e. those that pass through the body of the Earth
  - P waves are primary or push-pull waves
  - S waves are secondary, shear or shake waves



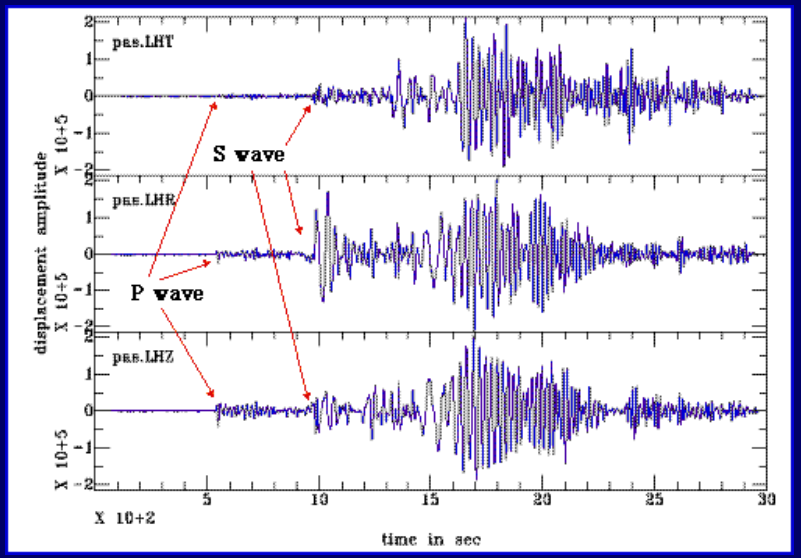
- P waves travel about 70% faster than S waves

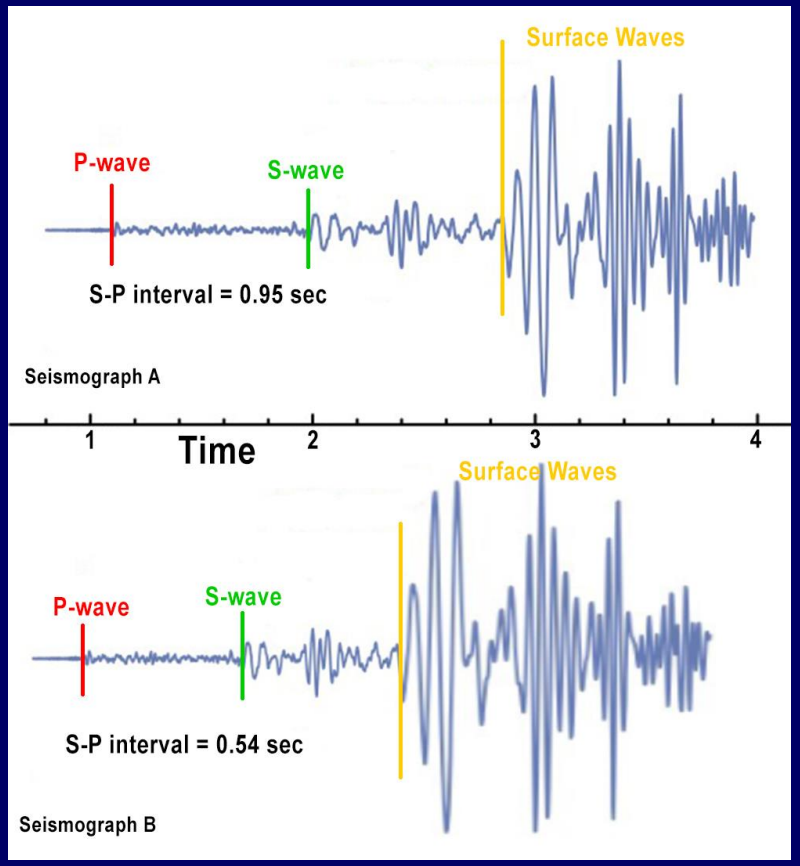
Seismology

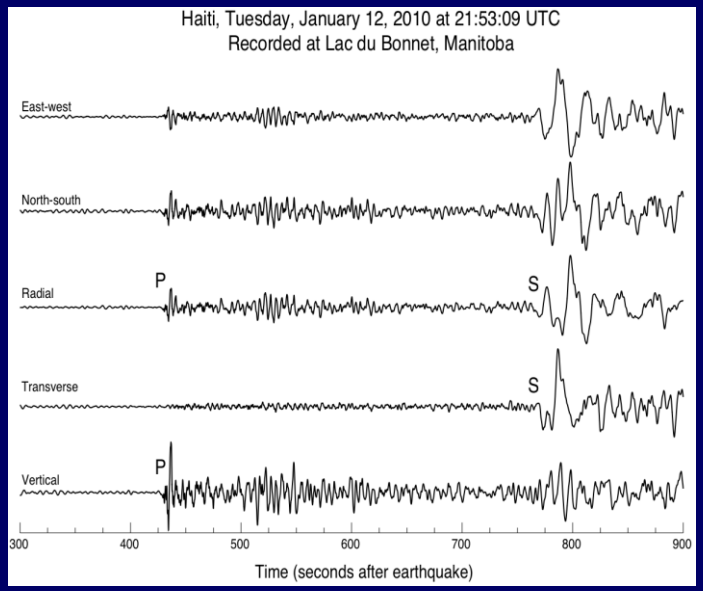


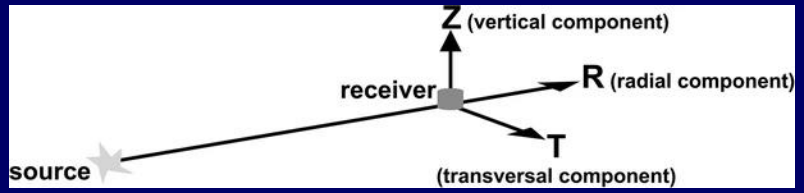
Seismograms provide information on **location**, depth and magnitude of an earthquake

Examples of recognizing P- and S-wave arrival

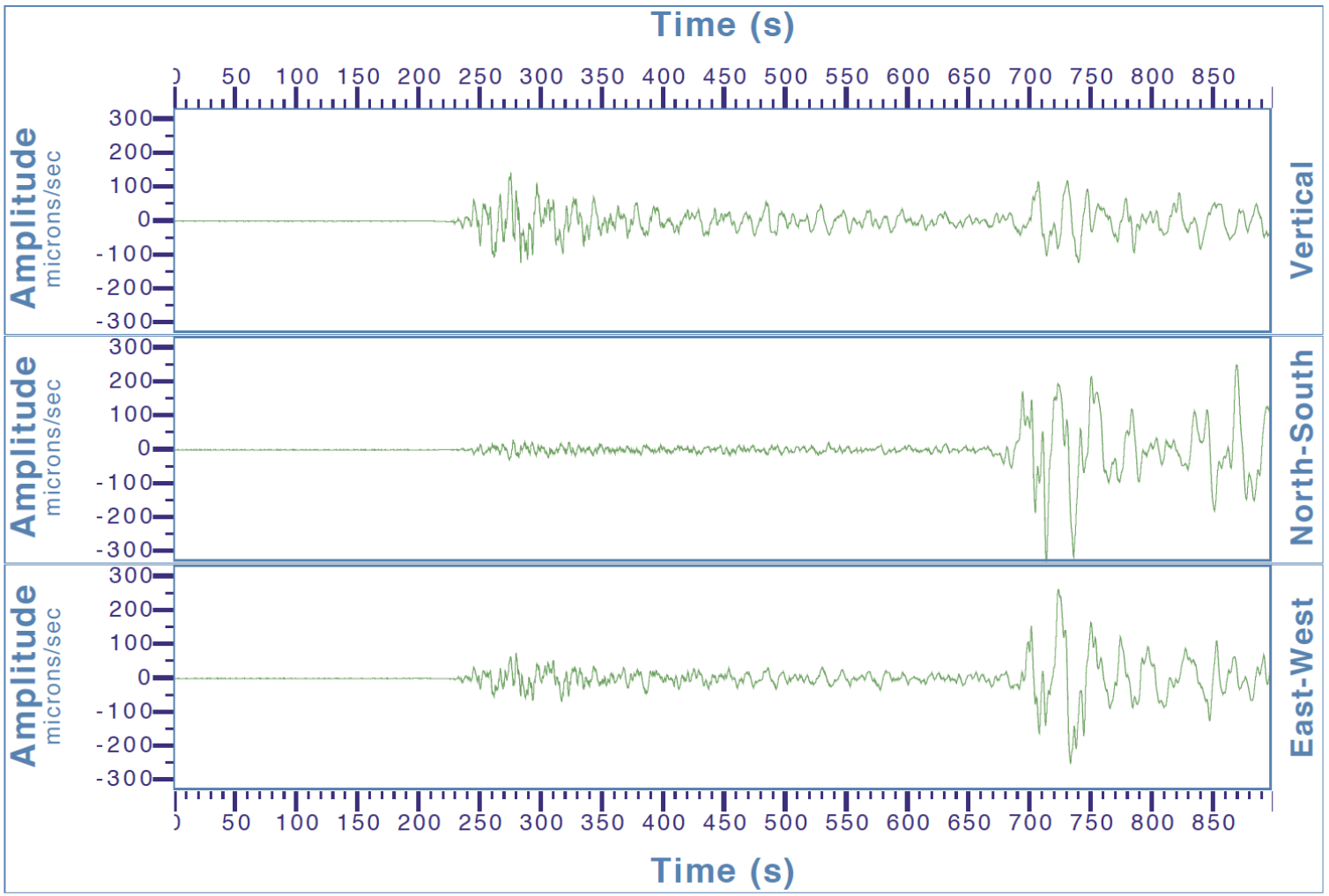






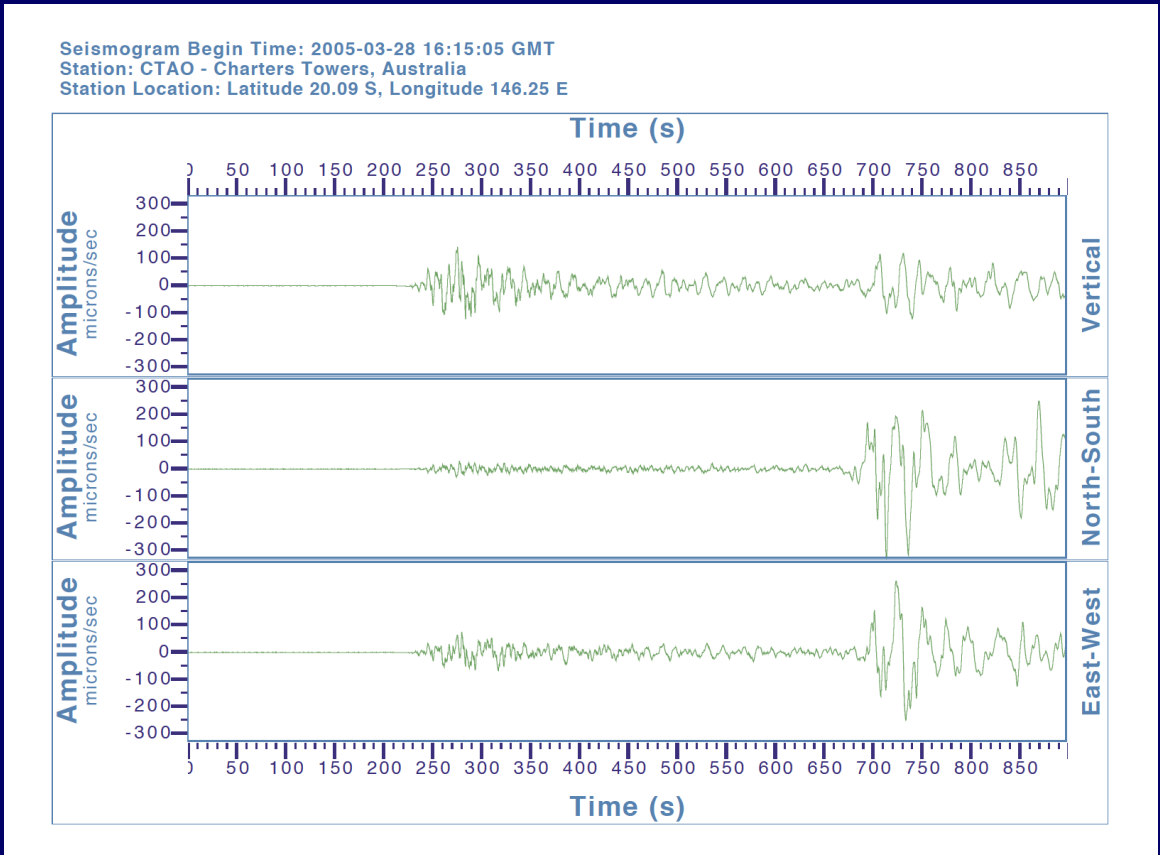


28 March 2005 N. Sumatra M 8.7 earthquake

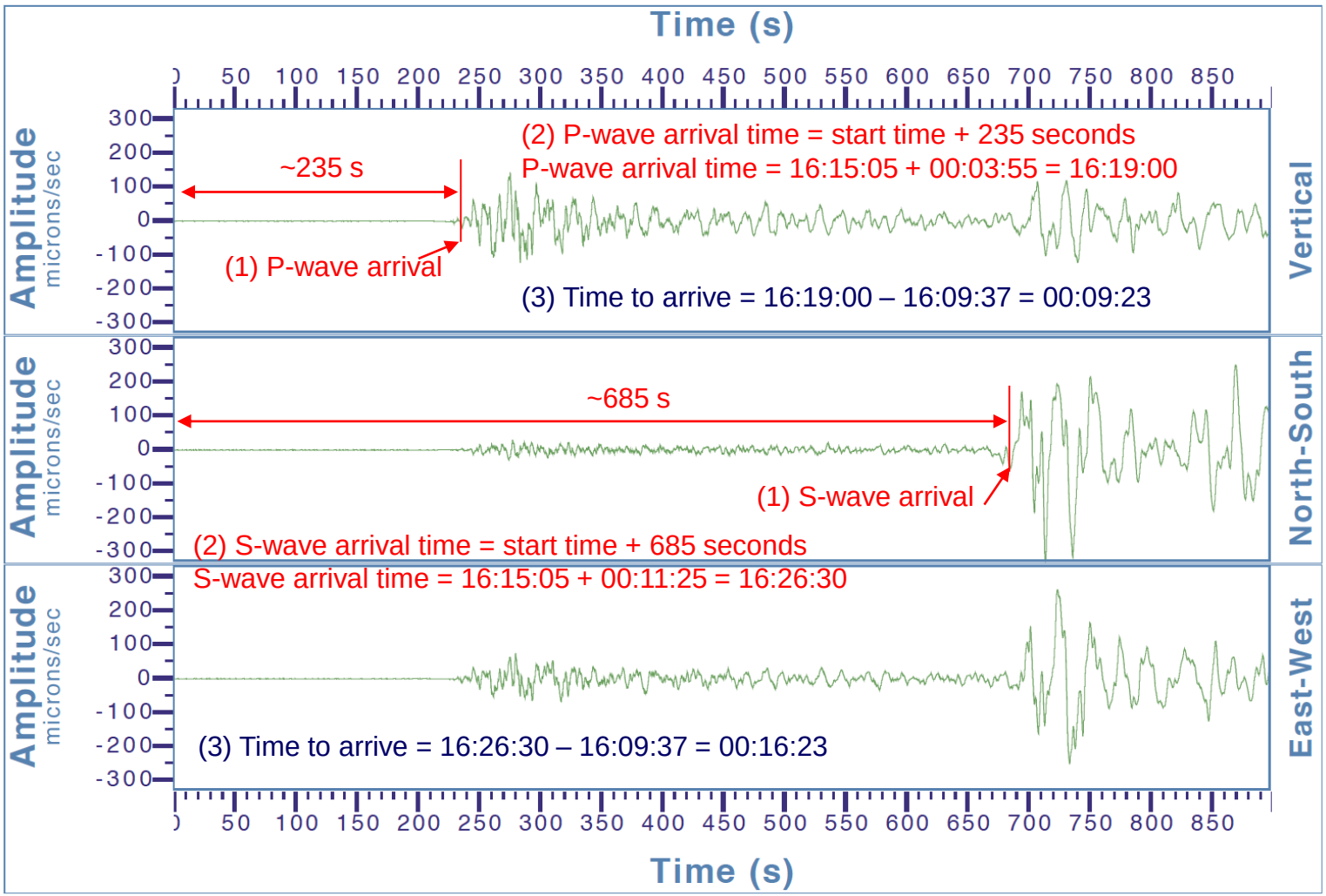


Tasks

1. Identify the P- and S-wave arrivals on the seismogram
2. Determine the arrival time of the P- and S-waves
3. Determine how long it took each wave to arrive at Charters Towers
4. In which component is most of the energy for the P-wave? …. S-wave?



Tasks- solutions

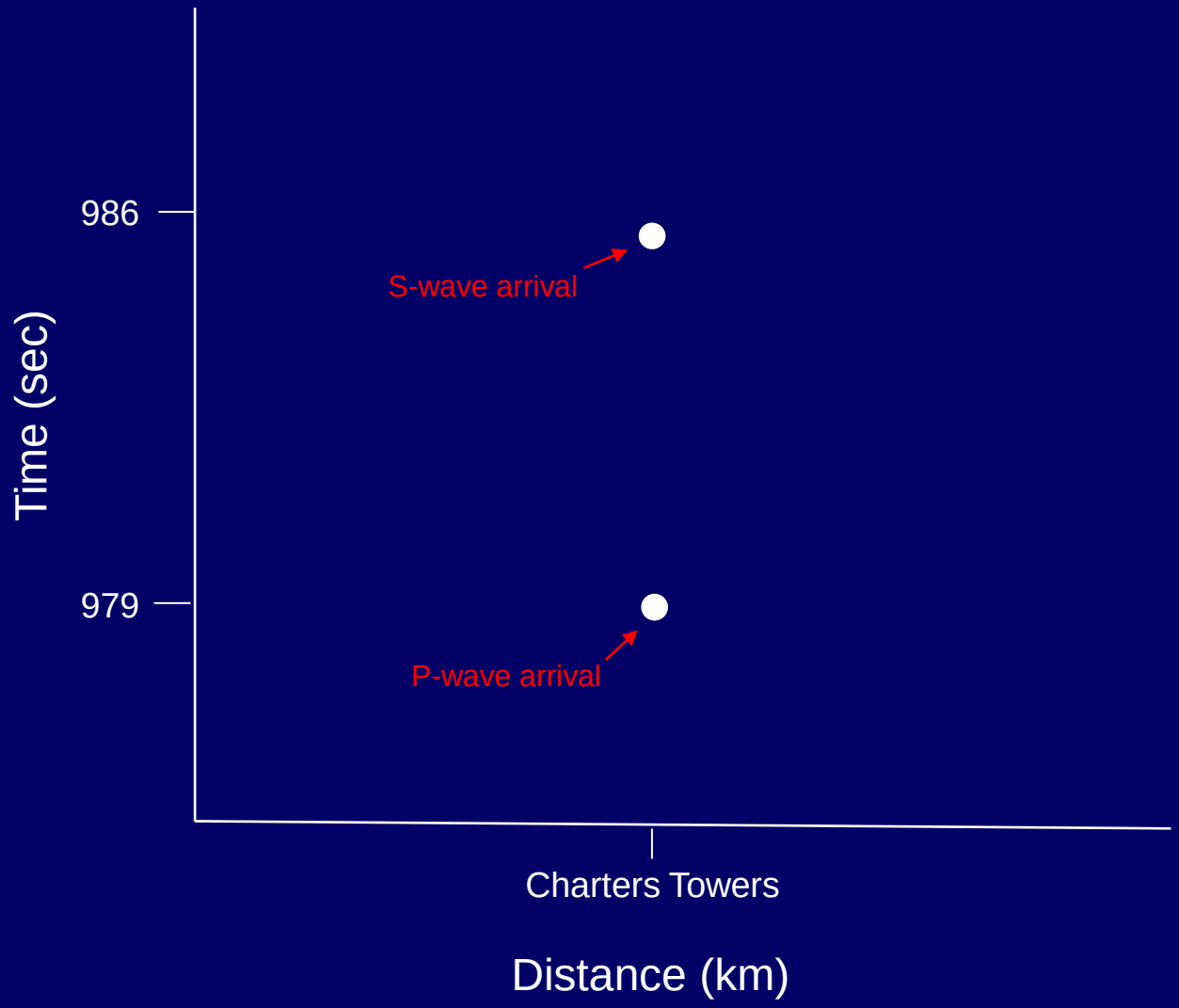


16 January 1995 Hyougo-ken Nanbu (Kobe) earthquake

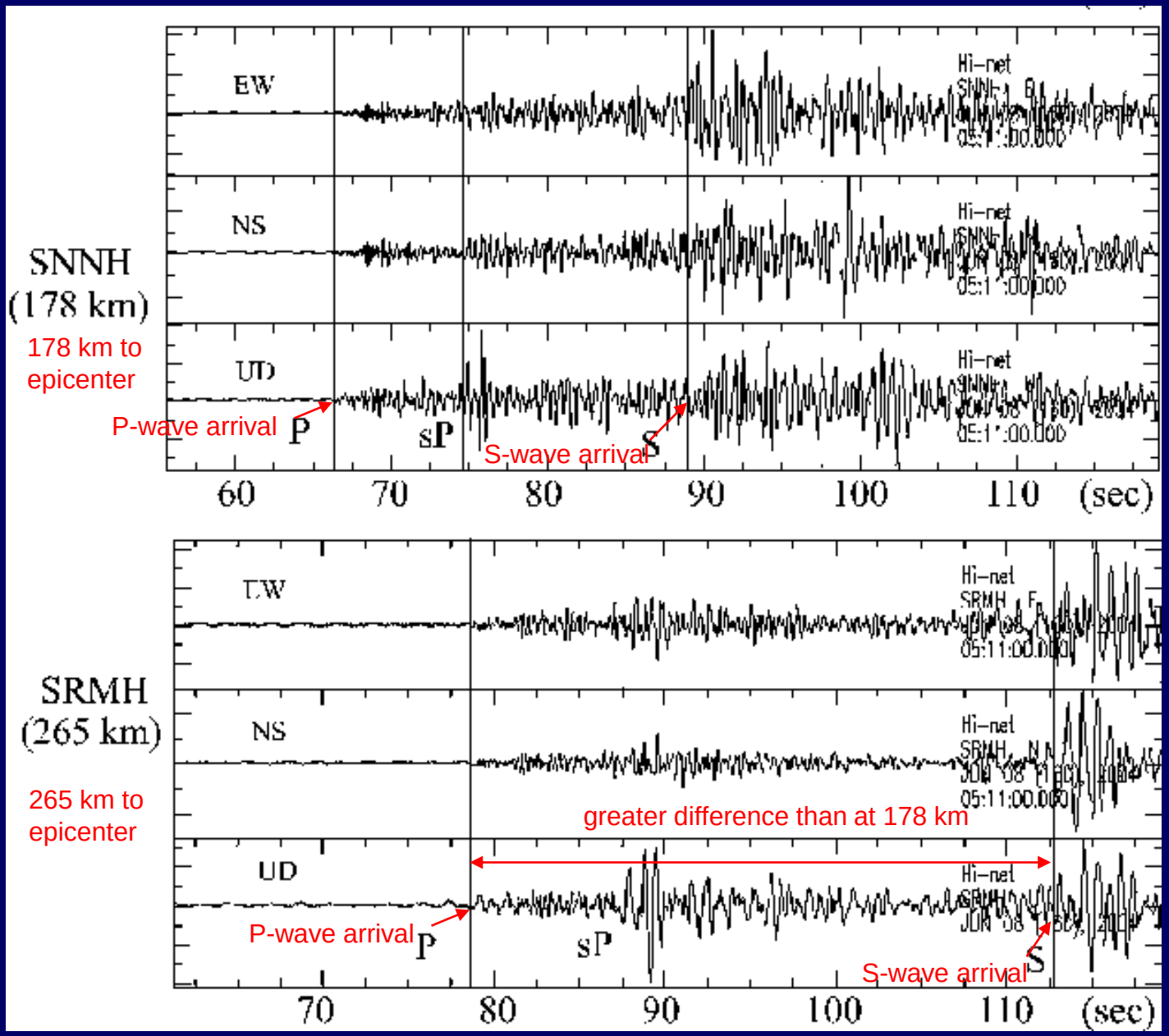
To help you understand the relationship between the seismograms and the ground motion, watch the video clip of the 1995 Kobe earthquake (7.2M)
http://www.youtube.com/watch?v=0plbf5w0sbA

Try and identify the arrival of the different seismic waves in the video and compare the magnitude and direction of the movement associated with the P- versus S-wave

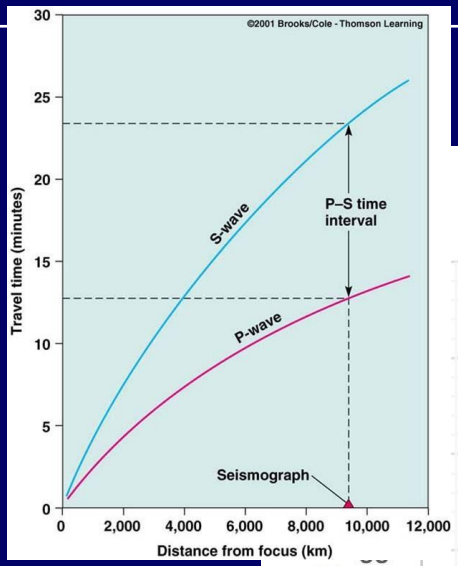
Calculating the travel time

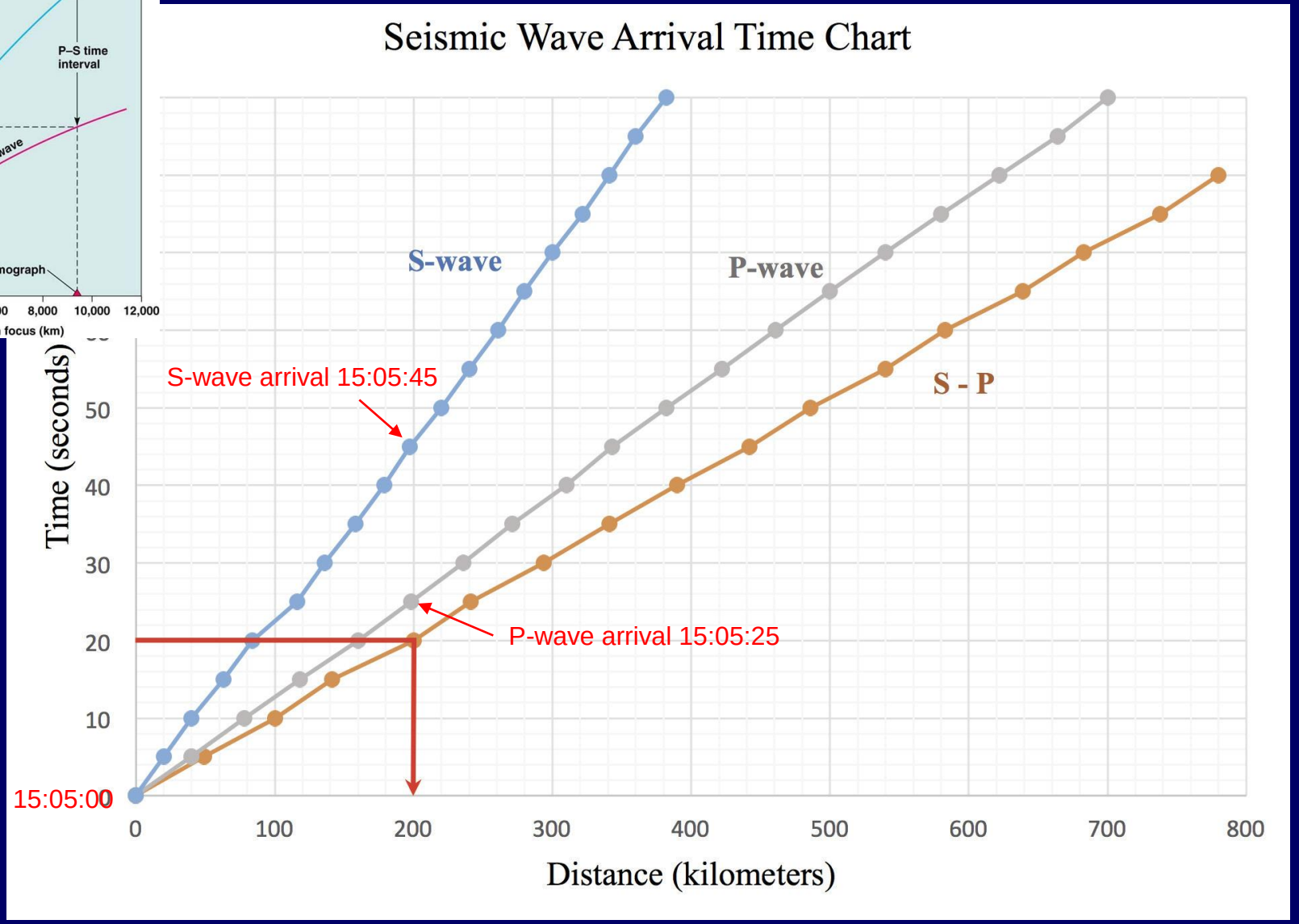


Difference in arrival time with distance



Calculating the travel time



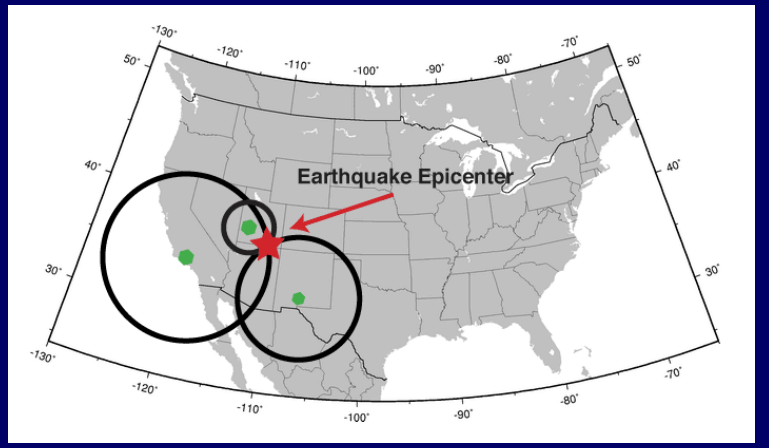


Practice 2 Tasks

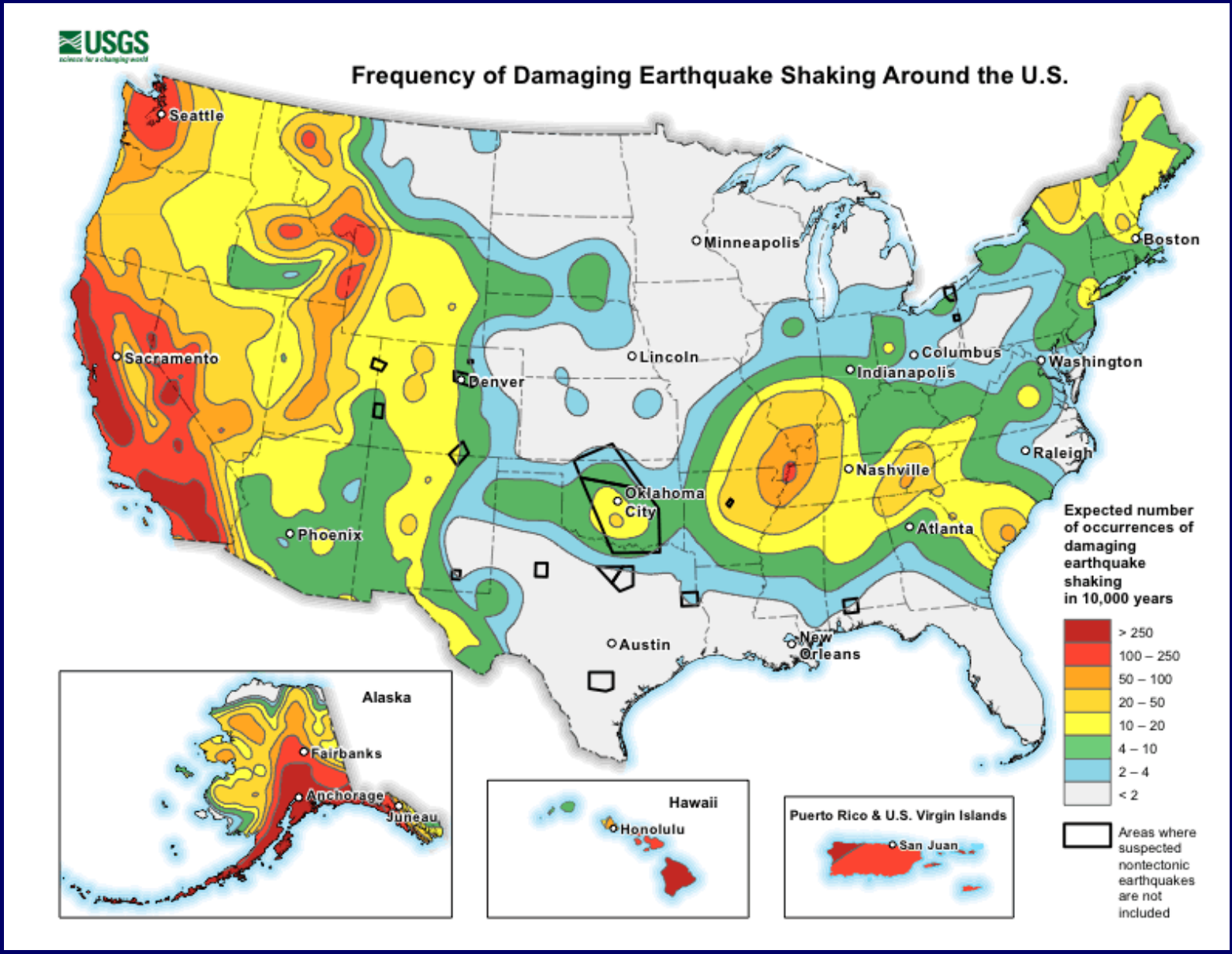
1. Identify the P- and S-wave arrivals on your 3 seismograph stations and note the difference in time between the two arrivals

2. Use the graph to estimate the distance between each station and the earthquake

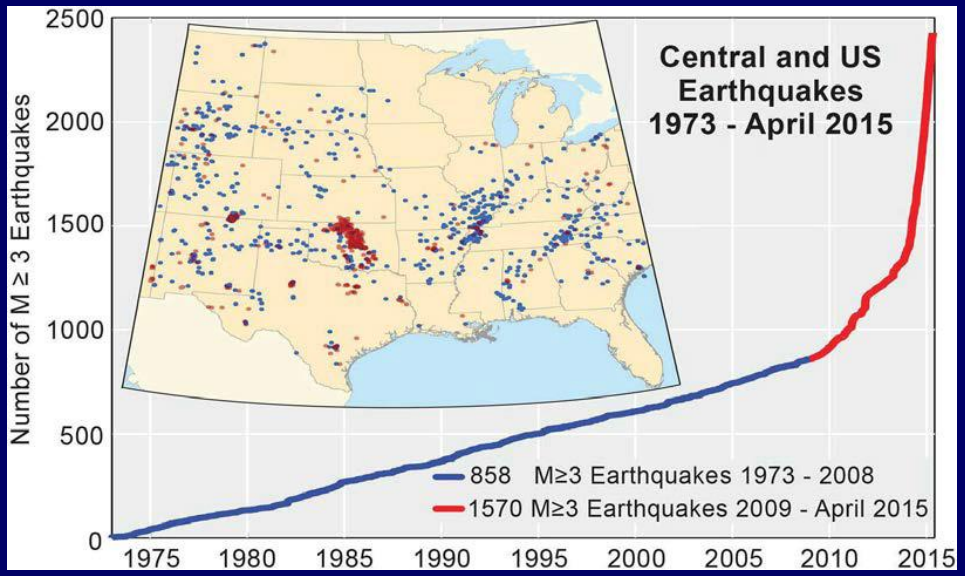
3. Use a triangulation method (see inset map below) to locate the region in which the epicenter of the earthquake is located.

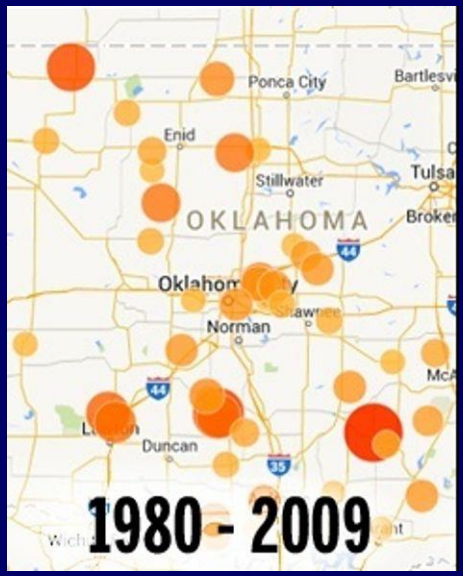


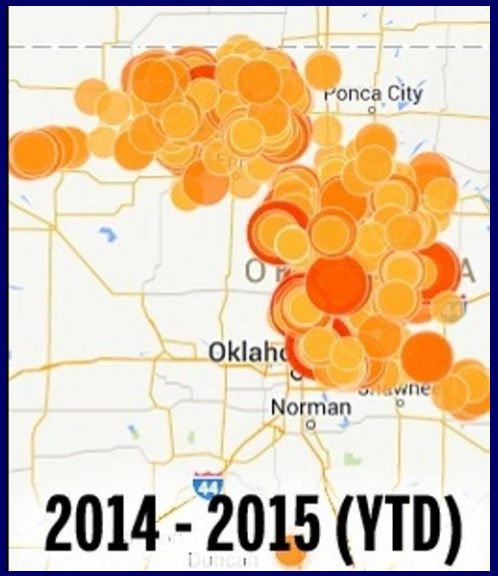
Earthquake distribution- USA

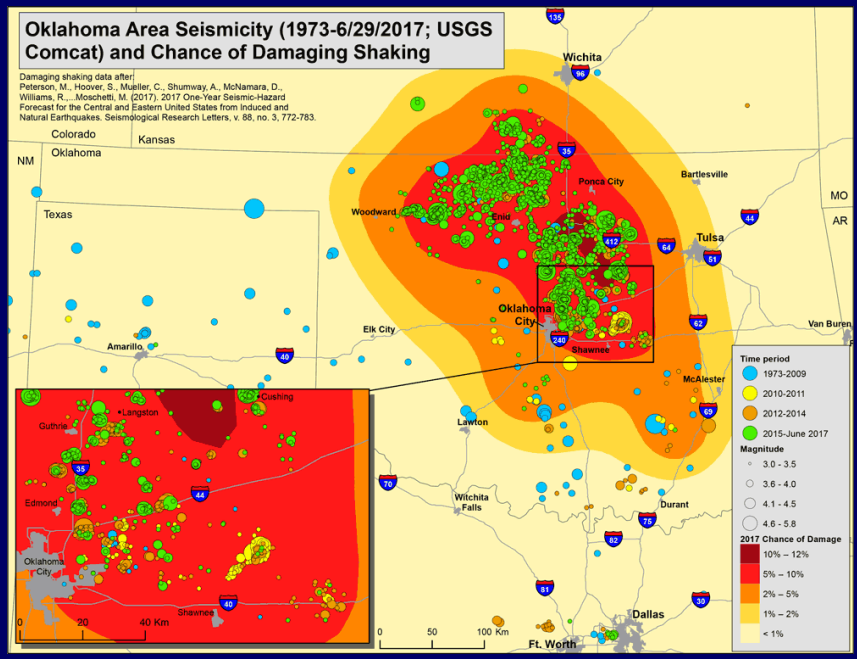


Induced earthquake activity









Asynchronous activity

Complete the following activities

- Practice 5 Assessing earthquake intensity
- Practice 6 Hazards: Liquefaction
- Practice 7 The Northern California Earthquake, April 18, 1906

Great websites for earthquake watchers

Link to geophysics institutes around the world

- https://www.usgs.gov/natural-hazards/earthquake-hazards/science/geophysics-institutions-throughout-world

The UK

- http://www.earthquakes.bgs.ac.uk

The USA

- https://earthquake.usgs.gov/earthquakes/map/?extent=8.88507,-135.70313&extent=59.46741,-54.31641&magnitude=all&settings=true